Leveraging G x E Interaction to Optimize Multi-Trait Selection in Lima Bean

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Authors

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---

```r
library(tinytex)
library(reshape2)
library(ggplot2)
library(openxlsx)
library(asreml)
```

```
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```

```r
library(dplyr)
library(tibble)
library(ggplot2)
library(metan)
library(writexl)
library(ggpubr)
```

Lima Bean Data

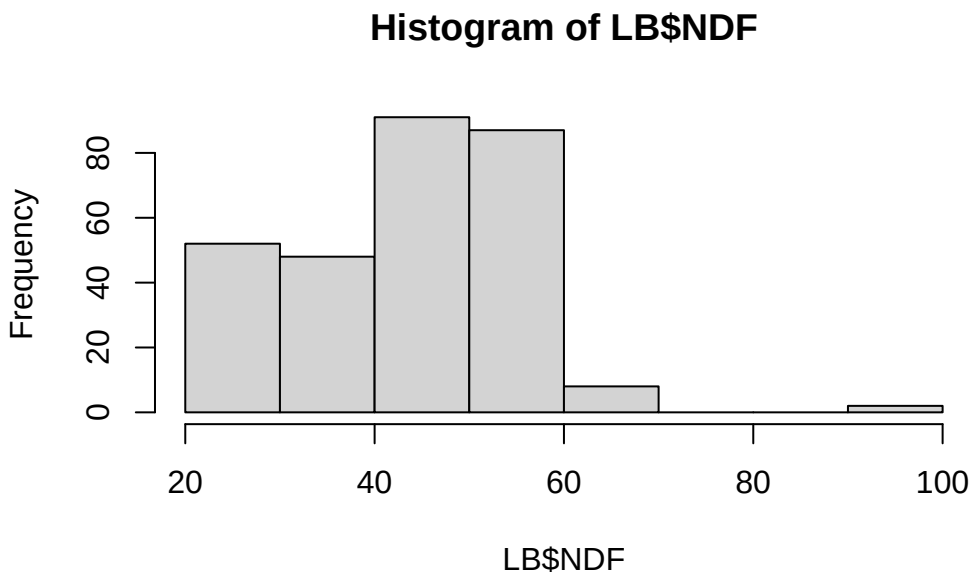
- 3 Locations (Brazil) - Piracicaba - SP, Teresina - PI, Tianguá - CE,
- 40 Lima Bean Lines
- Randomized Complete Block Design (RCBD) - 3 Repetitions
- Plot usable area: $5m^2$
- Traits:

Trait	Acronym	Unit
Grain Yield	GY	kg/ha
Number days to flowering	NDF	days
Number days to maturity	NDM	days
One Hundred Seed Weight	OHSW	g
Plant Height	PH	cm
Pod Length	PL	mm
Pod Number	PN	mm
Pod Thickness	PT	mm
Pod Width	PW	mm
Seed Length	SL	mm
Seed Thickness	ST	mm
Seed Width	SW	mm

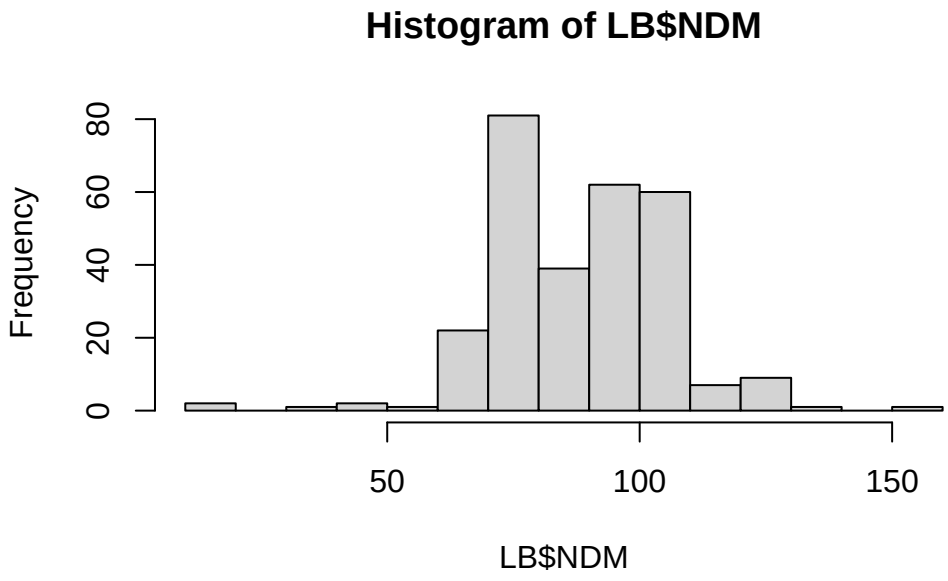
```r
LB = read.xlsx("C://Users//joaop//Desktop//fava//LimaBean.xlsx")
LB$Block    <- as.factor(LB$Block)
LB$Env    <- as.factor(LB$Env)
LB$Genotype <- as.factor(LB$Genotype)
LB$GEN<- as.factor(LB$GEN)
str(LB)
```

```
'data.frame':   288 obs. of  16 variables:
 $ Env     : Factor w/ 3 levels "Piracicaba","Teresina",..: 3 2 2 2 2 2 3 2 2 2 ...
 $ Block   : Factor w/ 3 levels "1","2","3": 2 1 1 1 2 1 2 1 1 3 ...
 $ Genotype: Factor w/ 40 levels "H25_53","H25_54",..: 37 39 16 11 28 38 25 28 40 38 ...
 $ GEN     : Factor w/ 40 levels "L01","L02","L03",..: 37 39 16 11 28 38 25 28 40 38 ...
 $ NDF     : num  30 36 63 42 91 39 30 64 52 48 ...
 $ NDM     : num  71 125 125 125 125 125 91 133 125 93 ...
 $ PH      : num  41.4 33.4 33.2 29.4 38.5 ...
 $ PL      : num  NA 55.3 55 51.5 48.6 ...
 $ PW      : num  NA 13.9 12.9 12.4 13.5 ...
 $ PT      : num  NA 7.96 8.81 6.72 7.19 8.69 9.15 7.42 6.12 8.28 ...
 $ SL      : num  NA 10.78 10.17 9.44 10.64 ...
 $ SW      : num  NA 6.82 6.62 6.2 7.37 7.8 9.01 7.28 7.6 7.48 ...
 $ ST      : num  NA 4.64 4.55 4.75 5.35 5.5 5.2 5.15 4.92 5.46 ...
 $ PN      : num  NA 22 5 37 35 32 25 48 62 62 ...
 $ GY      : num  NA 15.3 21.4 23.1 24.5 ...
 $ OHSW    : num  NA 24.9 10.7 26.5 28.8 35.5 46 25.2 27.1 35.5 ...
```

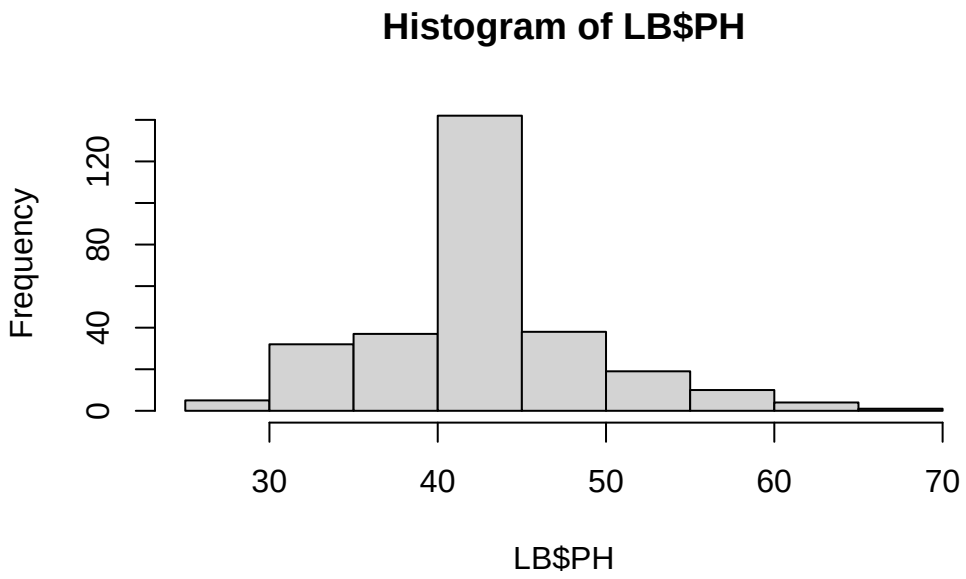
```r
hist(LB$NDF)
```


Histogram of LB$NDF
Frequency
0
20
40
60
80
20
40
60
80
100
LB$NDF

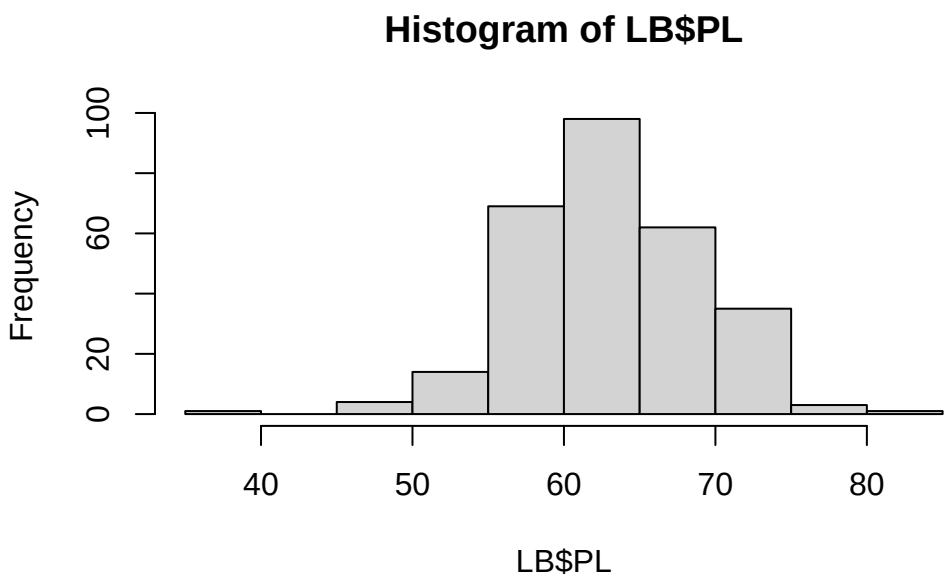
```r
hist(LB$NDM)
```


Histogram of LB$NDM
Frequency
0
20
40
60
80
50
100
150
LB$NDM

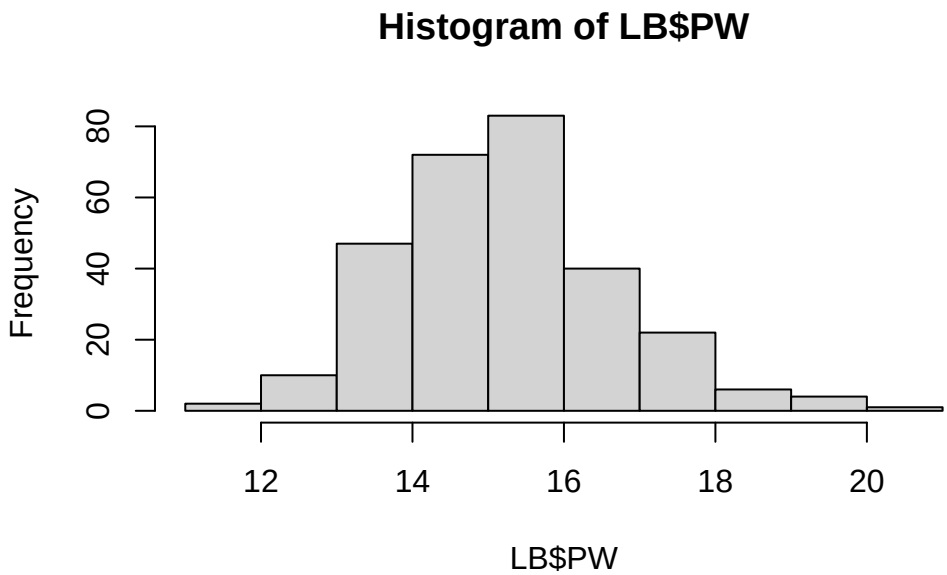
```r
hist(LB$PH)
```



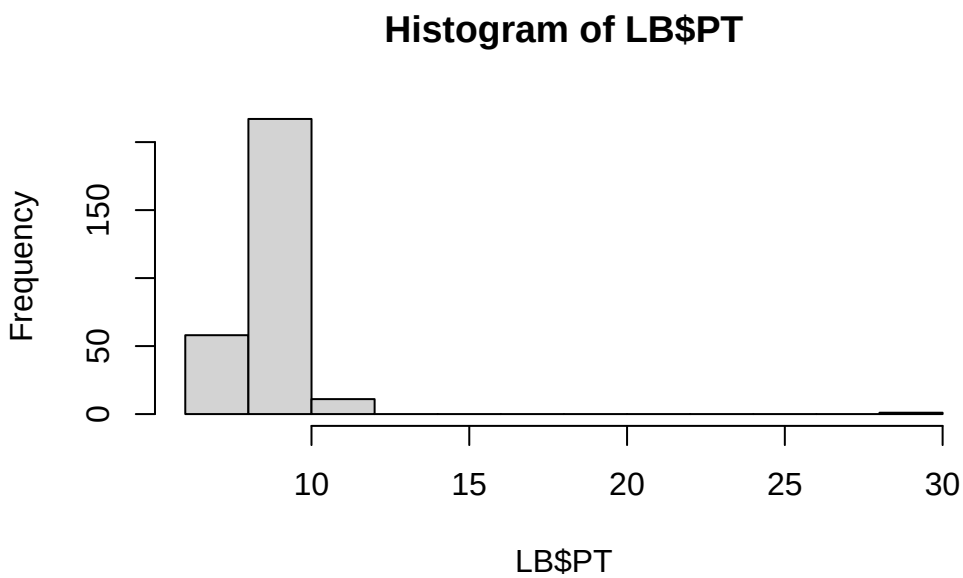

```r
hist(LB$PL)
```



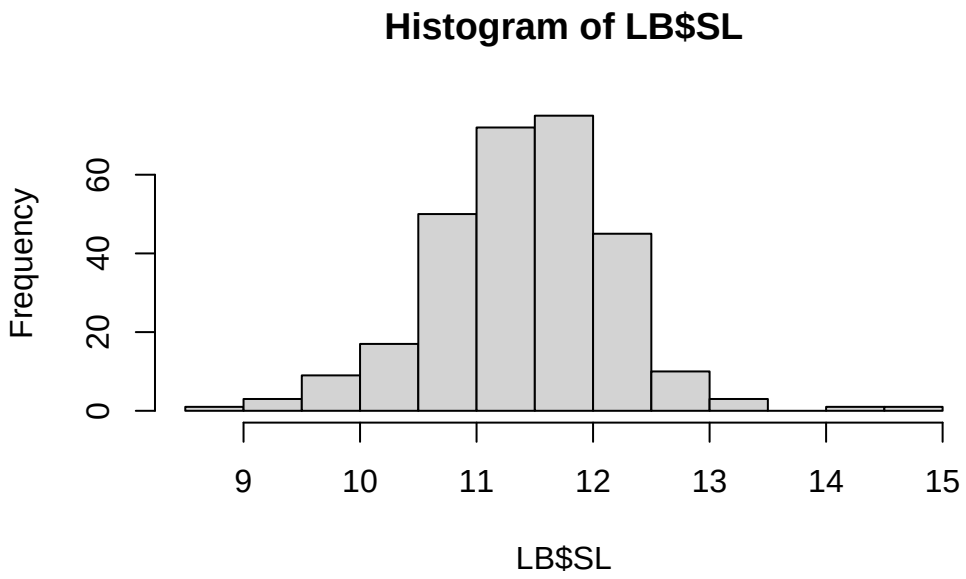
```r
hist(LB$PW)
```



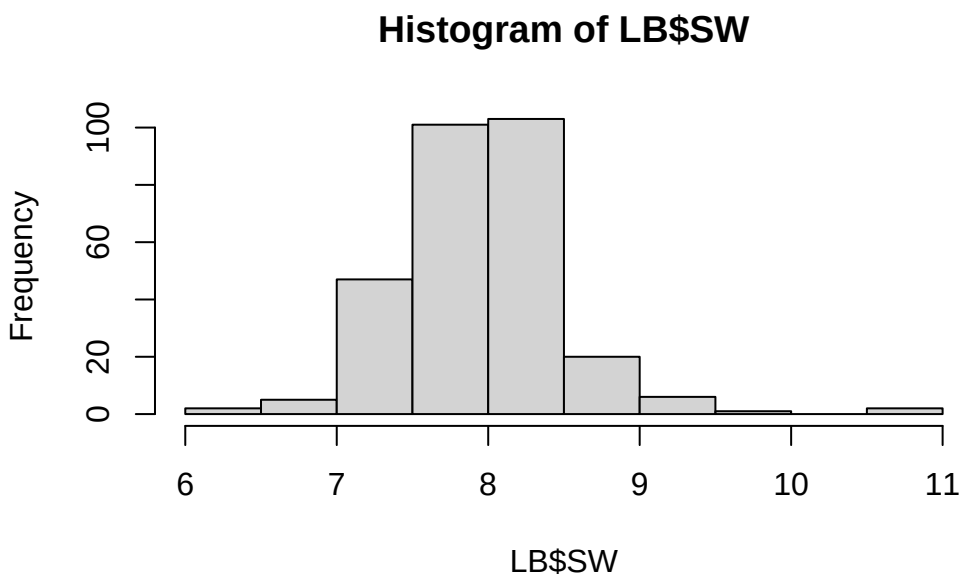

```r
hist(LB$PT)
```



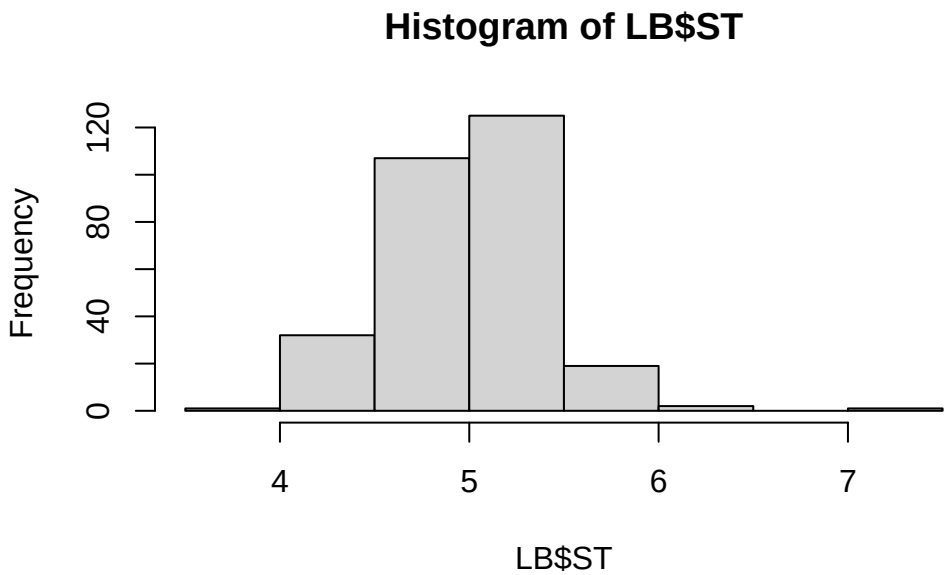
```r
hist(LB$SL)
```



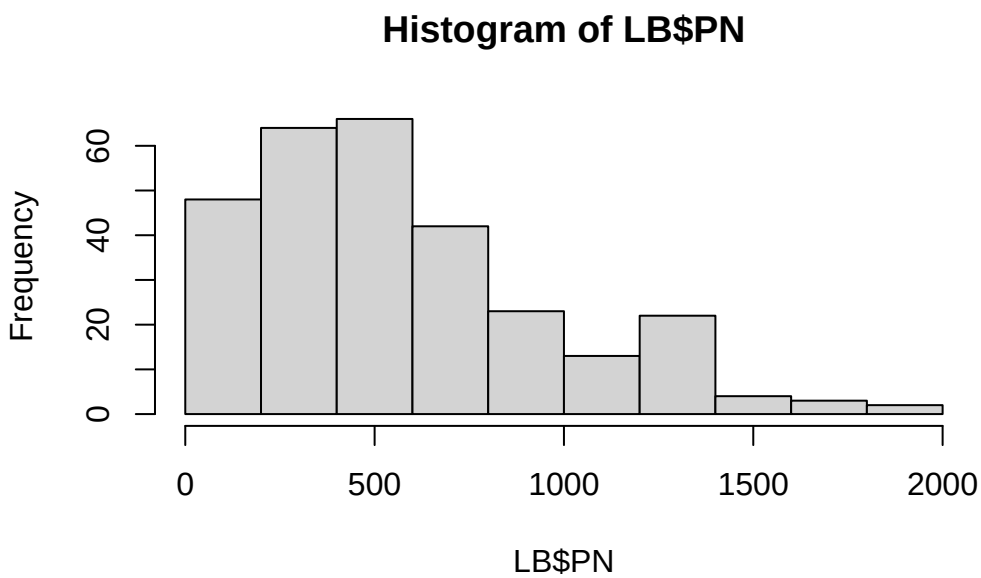

```r
hist(LB$SW)
```



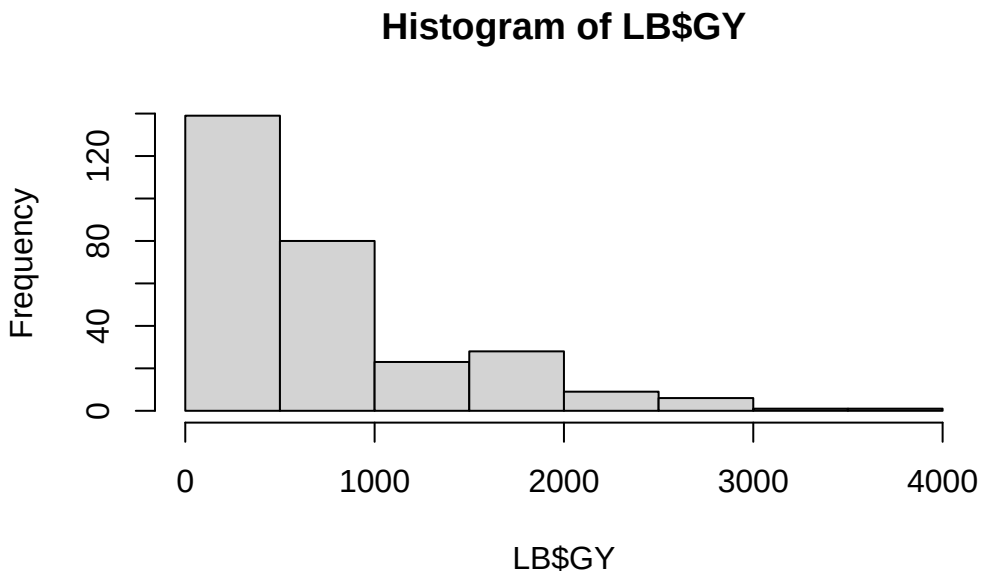
```r
hist(LB$ST)
```



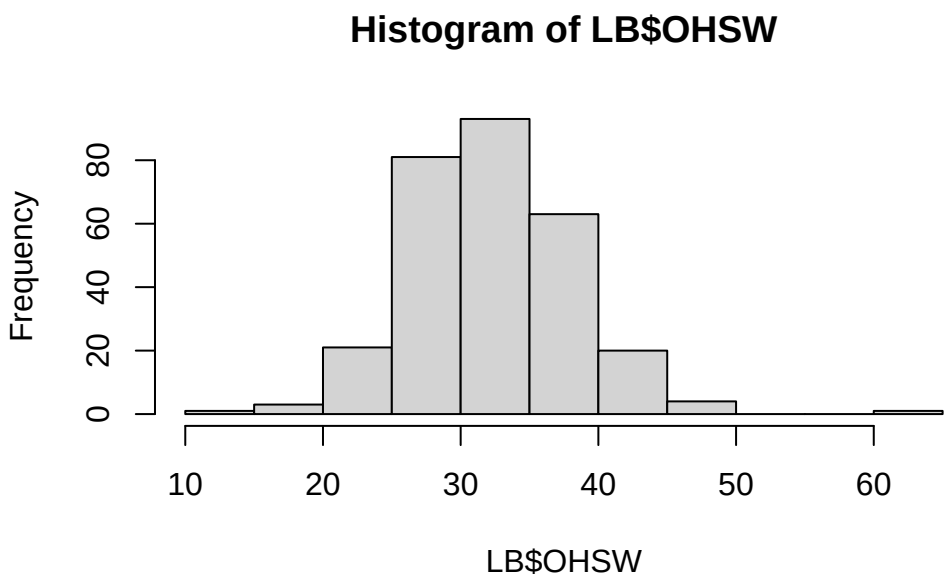

```r
hist(LB$PN)
```



```r
hist(LB$GY)
```




```r
hist(LB$OHSW)
```



anova

```
aov = anova_joint(LB,env = Env, gen = GEN, rep = Block,resp = everything())
```

Mixed-effect models

```
mixedmodel <- gamem_met(LB,env = Env, gen = GEN, rep = Block,random = "all",resp = everything
```

```
Evaluating trait NDF |====                                      | 8% 00:00:00
Evaluating trait NDM |=======                                   | 17% 00:00:01
Evaluating trait PH |===========                                | 25% 00:00:02

Evaluating trait PL |===============                            | 33% 00:00:03

Evaluating trait PW |==================                         | 42% 00:00:03

Evaluating trait PT |======================                     | 50% 00:00:04

Evaluating trait SL |==========================                 | 58% 00:00:05

Evaluating trait SW |=============================               | 67% 00:00:05

Evaluating trait ST |==================================          | 75% 00:00:06

Evaluating trait PN |=====================================       | 83% 00:00:07

Evaluating trait GY |========================================    | 92% 00:00:07

Evaluating trait OHSW |==========================================| 100% 00:00:08
```

```
-------------------------------------------------------------------------------
P-values for Likelihood Ratio Test of the analyzed traits
-------------------------------------------------------------------------------
    model      NDF      NDM       PH       PL       PW       PT       SL
 COMPLETE       NA       NA       NA       NA       NA       NA       NA
      GEN 2.58e-06 0.072308 4.25e-01 0.349315 3.88e-05 0.000875 5.79e-06
 REP(ENV) 6.99e-03 0.258106 1.06e-05 0.000616 2.07e-01 0.495762 4.42e-03
      ENV 9.09e-04 0.000451 7.35e-04 0.000311 1.00e-04 0.295555 4.72e-01
  GEN:ENV 8.70e-06 0.012307 3.12e-07 0.001036 2.96e-05 1.000000 1.66e-02
       SW      ST       PN       GY     OHSW
       NA      NA       NA       NA       NA
 1.94e-04 0.00077 2.63e-01 2.17e-01 3.82e-07
 1.46e-06 0.01363 2.21e-07 1.65e-09 6.56e-05
 2.14e-01 0.05708 6.54e-02 2.36e-02 1.36e-02
 9.20e-04 0.45230 5.16e-05 1.08e-03 3.39e-02
-------------------------------------------------------------------------------
Variables with nonsignificant GxE interaction
PT ST
-------------------------------------------------------------------------------
```

LRT

```r
GYLRT = mixedmodel$GY$LRT
OHSWLRT = mixedmodel$OHSW$LRT
NDFLRT = mixedmodel$NDF$LRT
NDMLRT = mixedmodel$NDM$LRT
PHLRT = mixedmodel$PH$LRT
PNLRT = mixedmodel$PN$LRT
LRT = list(
  GY_LRT   = GYLRT,
  OHSW_LRT = OHSWLRT,
  NDF_LRT  = NDFLRT,
  NDM_LRT  = NDMLRT,
  PH_LRT   = PHLRT,
  PN_LRT   = PNLRT
)
LRT
```

```
$GY_LRT
              model npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>            1    6 -2190.3 4392.7
(1 | GEN)         3    5 -2191.1 4392.2  1.526  1   0.216753
```

```
(1 | REP:ENV)     5    5 -2208.5 4427.0 36.350  1  1.648e-09 ***
(1 | ENV)         2    5 -2192.9 4395.8  5.123  1   0.023604 *
(1 | GEN:ENV)     4    5 -2195.7 4401.4 10.690  1   0.001077 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$OHSW_LRT
              model npar  logLik    AIC     LRT Df Pr(>Chisq)
<none>            1    6 -841.49 1695.0
(1 | GEN)         3    5 -854.38 1718.8 25.7834  1  3.820e-07 ***
(1 | REP:ENV)     5    5 -849.45 1708.9 15.9339  1  6.559e-05 ***
(1 | ENV)         2    5 -844.53 1699.1  6.0885  1    0.01361 *
(1 | GEN:ENV)     4    5 -843.74 1697.5  4.4999  1    0.03390 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$NDF_LRT
              model npar   logLik    AIC     LRT Df Pr(>Chisq)
<none>            1    6  -995.79 2003.6
(1 | GEN)         3    5 -1006.84 2023.7 22.1037  1  2.583e-06 ***
(1 | REP:ENV)     5    5  -999.43 2008.8  7.2745  1  0.0069941 **
(1 | ENV)         2    5 -1001.29 2012.6 11.0038  1  0.0009093 ***
(1 | GEN:ENV)     4    5 -1005.68 2021.3 19.7772  1  8.702e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$NDM_LRT
              model npar  logLik    AIC     LRT Df Pr(>Chisq)
<none>            1    6 -1173.2 2358.4
(1 | GEN)         3    5 -1174.8 2359.7  3.2298  1   0.072308 .
(1 | REP:ENV)     5    5 -1173.9 2357.7  1.2789  1   0.258106
(1 | ENV)         2    5 -1179.4 2368.8 12.3079  1   0.000451 ***
(1 | GEN:ENV)     4    5 -1176.4 2362.7  6.2661  1   0.012307 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$PH_LRT
              model npar  logLik    AIC     LRT Df Pr(>Chisq)
<none>            1    6 -831.83 1675.7
(1 | GEN)         3    5 -832.15 1674.3  0.6359  1  0.4252042
(1 | REP:ENV)     5    5 -841.53 1693.0 19.3926  1  1.064e-05 ***
(1 | ENV)         2    5 -837.53 1685.1 11.3988  1  0.0007349 ***
(1 | GEN:ENV)     4    5 -844.92 1699.8 26.1735  1  3.121e-07 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$PN_LRT
              model npar  logLik    AIC     LRT Df Pr(>Chisq)
<none>            1    6 -2063.9 4139.8
(1 | GEN)         3    5 -2064.5 4139.0  1.2548  1    0.26264
(1 | REP:ENV)     5    5 -2077.3 4164.6 26.8382  1  2.212e-07 ***
(1 | ENV)         2    5 -2065.6 4141.1  3.3947  1    0.06541 .
(1 | GEN:ENV)     4    5 -2072.1 4154.1 16.3888  1  5.159e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
write_xlsx(LRT, "LRT_LB.xlsx")
```

BLUPs

```r
BLUPgen = list(
  GY_BLUP   = mixedmodel$GY$BLUPgen,
  OHSW_BLUP = mixedmodel$OHSW$BLUPgen,
  NDF_BLUP  = mixedmodel$NDF$BLUPgen,
  NDM_BLUP  = mixedmodel$NDM$BLUPgen,
  PH_BLUP   = mixedmodel$PH$BLUPgen,
  PN_BLUP   = mixedmodel$PN$BLUPgen
)
```

```r
write_xlsx(BLUPgen, "BLUPgen_LB.xlsx")
```

```r
library(dplyr)
library(tibble)

ebvMat <- data.frame(
  GEN = mixedmodel$GY$BLUPgen$GEN,
  GY   = mixedmodel$GY$BLUPgen$Predicted,
  OHSW = mixedmodel$OHSW$BLUPgen$Predicted,
  NDF  = mixedmodel$NDF$BLUPgen$Predicted,
  NDM  = mixedmodel$NDM$BLUPgen$Predicted,
  PH   = mixedmodel$PH$BLUPgen$Predicted,
  PN   = mixedmodel$PN$BLUPgen$Predicted
) |>
  column_to_rownames("GEN") |>
```

```r
  as.matrix()
dim(ebvMat)
```

```
[1] 40  6
```

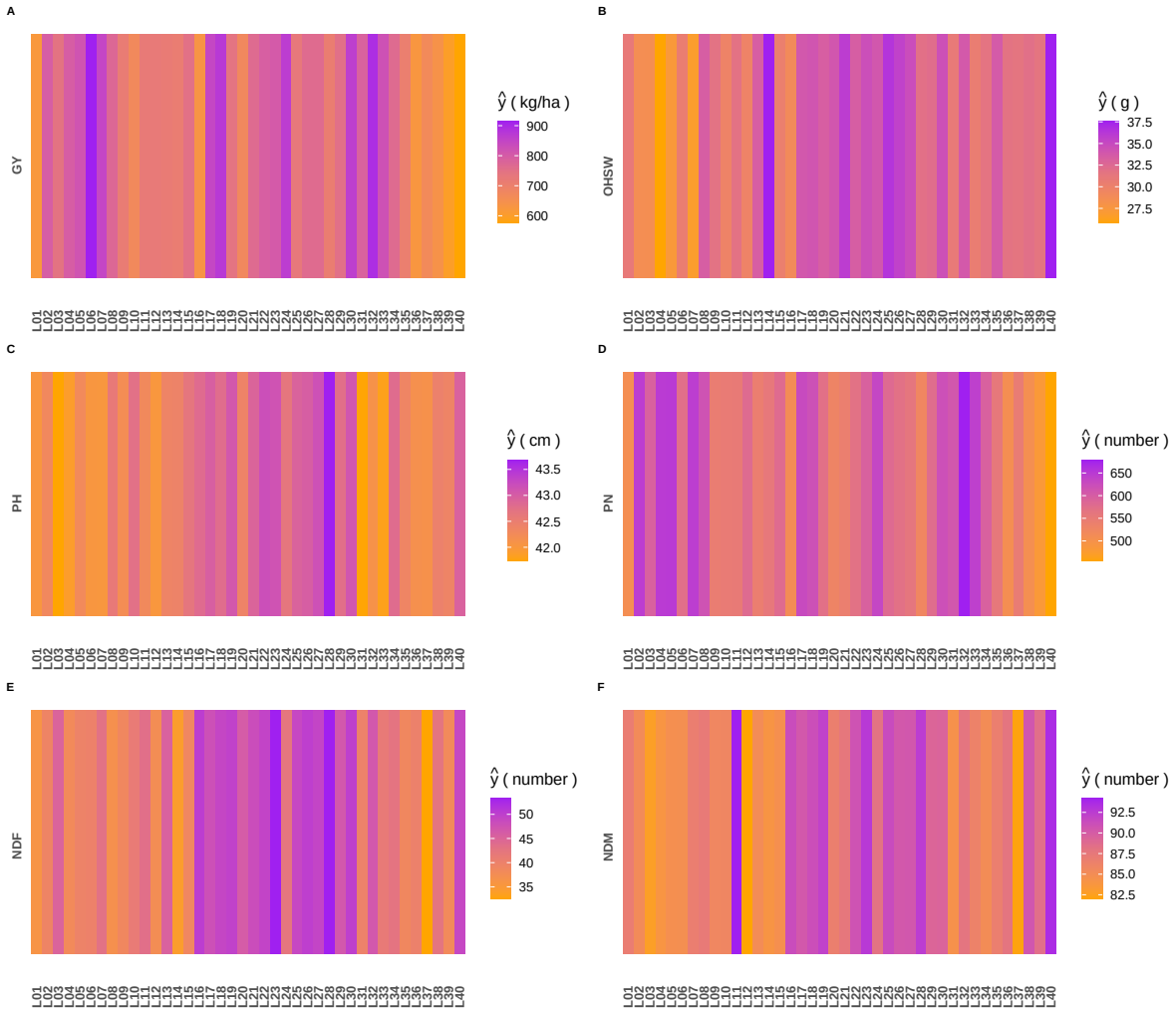
```r
head(ebvMat)
```

```
          GY     OHSW      NDF      NDM       PH       PN
L06 912.9283 37.51149 53.38240 94.24674 43.68422 679.0416
L32 892.1046 37.50313 53.38240 93.57356 43.18964 649.7793
L18 870.5821 36.35293 50.33583 92.73047 43.16320 648.9372
L24 863.9096 35.79919 50.06701 92.46701 43.13106 645.9057
L30 860.4754 35.47124 49.97741 92.09816 43.09918 645.6447
L07 845.4605 34.65497 49.70859 91.41315 43.07507 642.6915
```

```r
trait_units <- c(
  GY   = "kg/ha",   # kg·ha⁻¹
  OHSW = "g",
  PH   = "cm",
  PN   = "number",
  NDF  = "number",
  NDM  = "number"
)
plot_trait_heatmap <- function(x, trait_name) {

  gen_order <- sprintf("L%02d", 1:40)

  df <- x[[trait_name]]$BLUPgen %>%
    filter(GEN %in% gen_order) %>%
    mutate(
      GEN = factor(GEN, levels = gen_order),
      Trait = trait_name
    )

  ggplot(df, aes(x = GEN, y = Trait, fill = Predicted)) +
    geom_tile() +
    scale_fill_gradientn(
      colours = c("orange", "#E37383", "purple"),
      name = bquote(hat(y)~"("~.(trait_units[[trait_name]])~")")
    ) +
    labs(x = NULL, y = NULL) +
```

```r
    theme_minimal(base_size = 14) +
    theme(
      axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, face = "bold"),
      axis.text.y = element_text(angle = 90, face = "bold"),
      legend.position = "right",
      panel.grid = element_blank()
    )
}
traits <- c("GY", "OHSW", "PH", "PN", "NDF", "NDM")

BLUP_heatmaps <- lapply(traits, function(trait) {
  plot_trait_heatmap(mixedmodel, trait)
})
```

```r
blups_heatmap_panel <- ggarrange(
  plotlist = BLUP_heatmaps,
  ncol = 2, nrow = 3,
  labels = c("A", "B", "C", "D", "E", "F"), font.label = list(size = 10, face = "bold"),
  align = "v"
)
blups_heatmap_panel
```

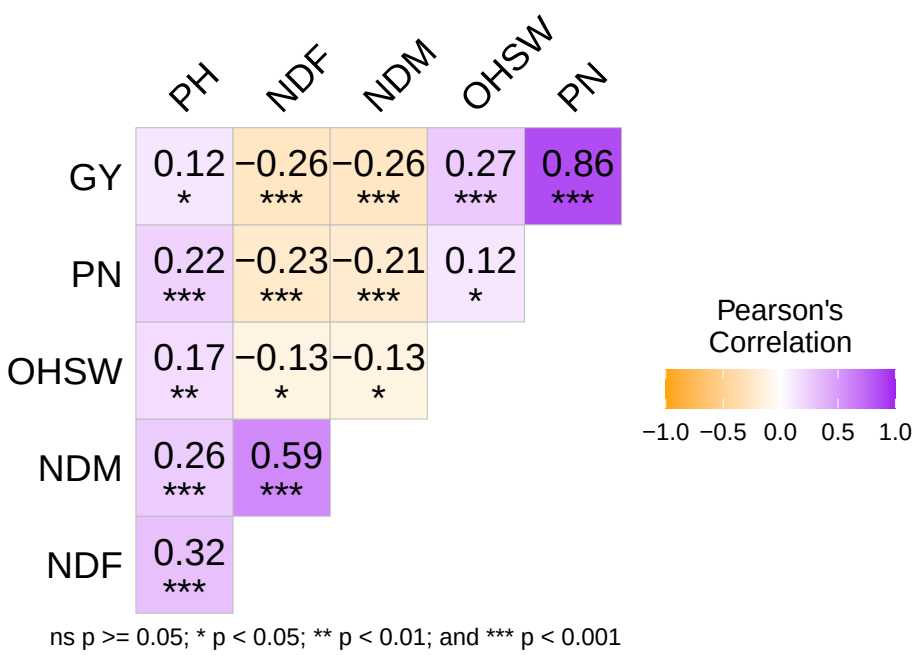



```
ggsave(
  "BLUPs_LB.pdf",
  blups_heatmap_panel,
  width = 14,
  height = 12,
  dpi = 600
)
```

Correlation

```
corrplot = plot(
  corr_coef(LB, NDF, NDM, PH, PN, GY, OHSW),
```

```
  type = "upper",
  diag = FALSE,
  reorder = T,
  signif = c("stars", "pval"),
  show = c("all", "signif"),
  p_val = 0.05,
  caption = TRUE,
  digits.cor = 2,
  digits.pval = 3,
  col.low = "orange",
  col.mid = "white",
  col.high = "purple",
  lab.x.position = "top",
  lab.y.position = "left",
  legend.position = "right",
  legend.title = "Pearson's\nCorrelation",
  size.text.cor = 5,
  size.text.signif = 5,
  size.text.lab = 14
)
corrplot
```



```
ggsave("Correlation.pdf",corrplot, width = 7, height = 6, dpi = 600)
```

Genotype by trait biplot

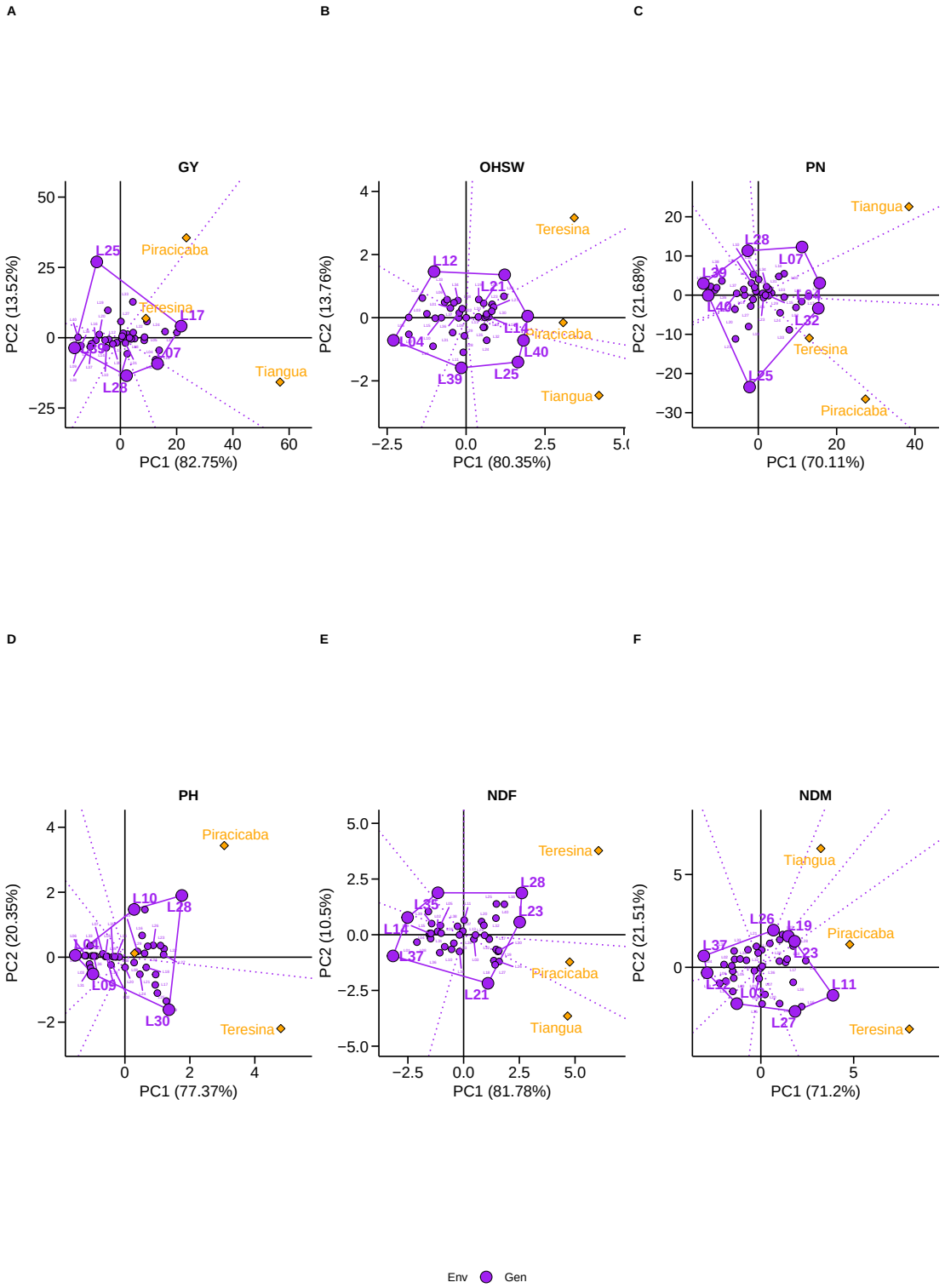
```
gtb1 = gtb(LB, GEN, resp = c(NDF, NDM, PH, PL, PW,
             PN, GY, OHSW))

gtb1plot = plot(gtb1,
     col.env = "orange",
            col.gen = "purple",
            size.text.gen = 5,
            size.text.env = 8, repel = T,
            repulsion = 1,
            col.line = "orange",
            plot_theme = theme_metan_minimal())
```

```
ggsave("Genotypebytraitbiplot_LB.pdf",gtb1plot, width = 8, height = 8, dpi = 600)
```

Which-won-Where

```
library(ggpubr)
library(ggplot2)

www <- list()

for(tr in traits){
  gge3_tr <- gge(LB, Env, GEN, resp = tr, svp = "symmetrical")

  p <- plot(gge3_tr, type = 3,
            col.env = "orange",
            col.gen = "purple",
            arrow.color = "purple",
            size.text.gen = 1,
            size.text.env = 4,
            size.shape.win = 4,
            large_label = 1,
            repel = TRUE,
            size.text.win = 4,
            repulsion = 0.5,
            col.line = "orange",
```

```r
            plot_theme = theme_metan_minimal())

  p_try <- try({
    p2 <- p + labs(title = tr, subtitle = NULL) +
            theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p <- p2
    NULL
  }, silent = TRUE)

  if(inherits(p_try, "try-error") || !inherits(p, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p)
    p <- p_grob + labs(title = tr, subtitle = NULL) +
              theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }

  www[[tr]] <- p
}

www_lb <- ggarrange(
  www[["GY"]],  www[["OHSW"]], www[["PN"]],
  www[["PH"]],  www[["NDF"]],  www[["NDM"]],
  ncol = 3, nrow = 2,
  common.legend = TRUE, legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(size = 10, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5
)

www_lb <- annotate_figure(
  www_lb,
  top = text_grob("", size = 14),
  fig.lab = "",
  fig.lab.pos = "bottom"
)
www_lb
```

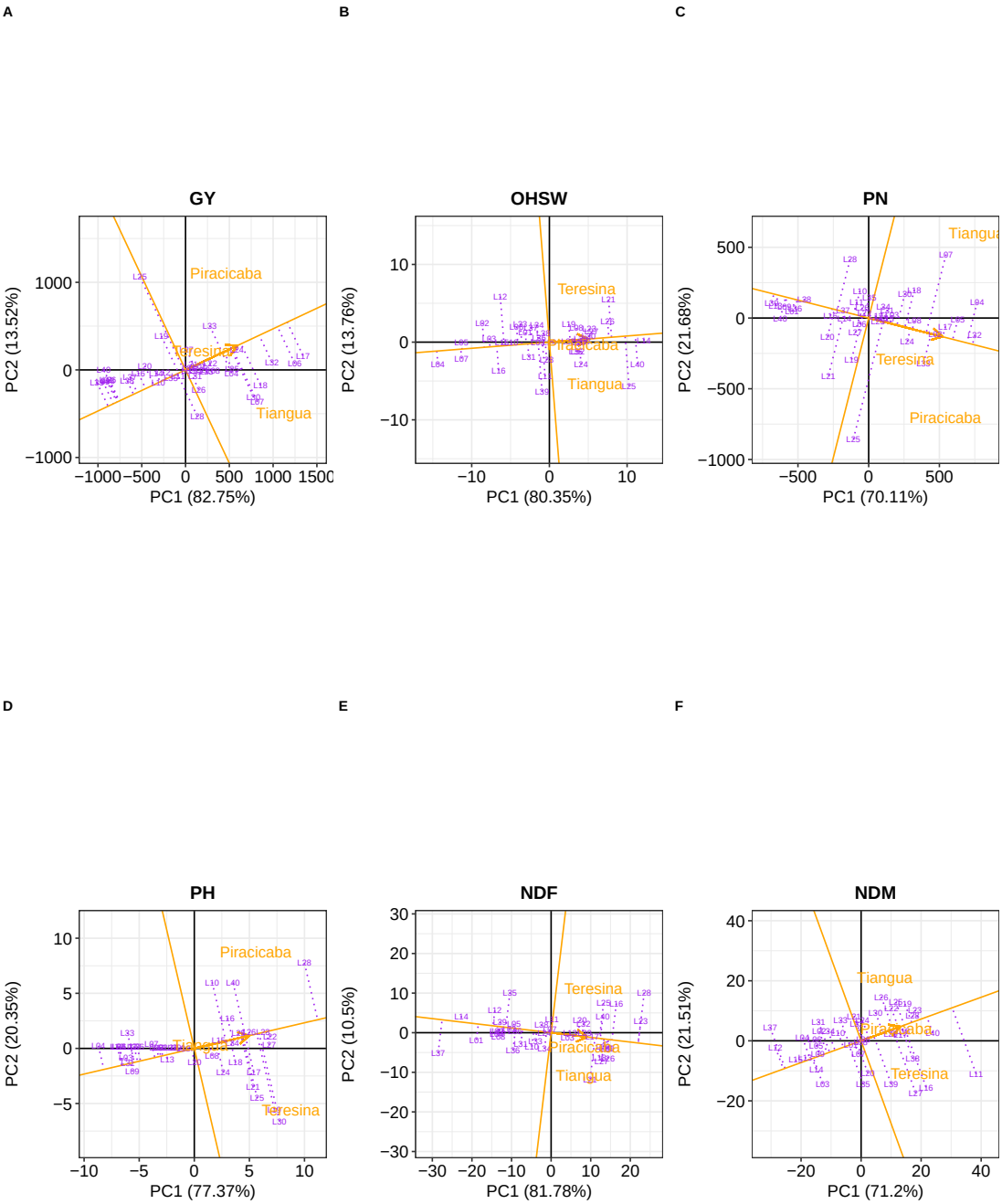

A
GY
PC2 (13.52%)
PC1 (82.75%)
Piracicaba
Teresina
Tiangua
L25
L17
L07
L28
B
OHSW
PC2 (13.76%)
PC1 (80.35%)
Teresina
Piracicaba
Tiangua
L12
L21
L14
L40
L04
L39
L25
C
PN
PC2 (21.68%)
PC1 (70.11%)
Tiangua
Teresina
Piracicaba
L28
L07
L32
L25
D
PH
PC2 (20.35%)
PC1 (77.37%)
Piracicaba
Teresina
L10
L28
L09
L30
E
NDF
PC2 (10.5%)
PC1 (81.78%)
Teresina
Piracicaba
Tiangua
L28
L35
23
L14
L37
L21
F
NDM
PC2 (21.51%)
PC1 (71.2%)
Tiangua
Piracicaba
Teresina
L26
L19
L37
L23
L11
L27
Env
Gen

```r
ggsave("Which-won-Where.pdf", www_lb, width = 10, height = 14, dpi = 600)
```

Mean_performance vs. stability

```r
dir.create("Mean_performance_vs_stability", showWarnings = FALSE)

mps_list <- list()

for(tr in traits){
    gge2_tr <- gge(
    LB,
    Env,
    GEN,
    resp = tr,
    svp = "genotype"
  )

  # Plot type 2
  p <- plot(gge2_tr,
            type = 2,
            col.env = "orange",
            col.gen = "purple",
            size.text.gen = 2,
            size.text.env = 4,
            repel = TRUE,
            repulsion = 2,
            col.line = "orange",
            plot_theme = theme_bw())

  p_try <- try({
    p2 <- p + labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p <- p2
    NULL
  }, silent = TRUE)

  if(inherits(p_try, "try-error") || !inherits(p, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p)
    p <- p_grob + labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }
```

```
  mps_list[[tr]] <- p2}

mps_panel <- ggarrange(
  mps_list[["GY"]],  mps_list[["OHSW"]], mps_list[["PN"]],
  mps_list[["PH"]],  mps_list[["NDF"]],  mps_list[["NDM"]],
  ncol = 3, nrow = 2,
  common.legend = TRUE, legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(size = 10, face = "bold"),
  align = "hv"
)

mps_panel = annotate_figure(
  mps_panel,
  top = text_grob("", size = 14),
  fig.lab = "",
  fig.lab.pos = "bottom"
)
mps_panel
```

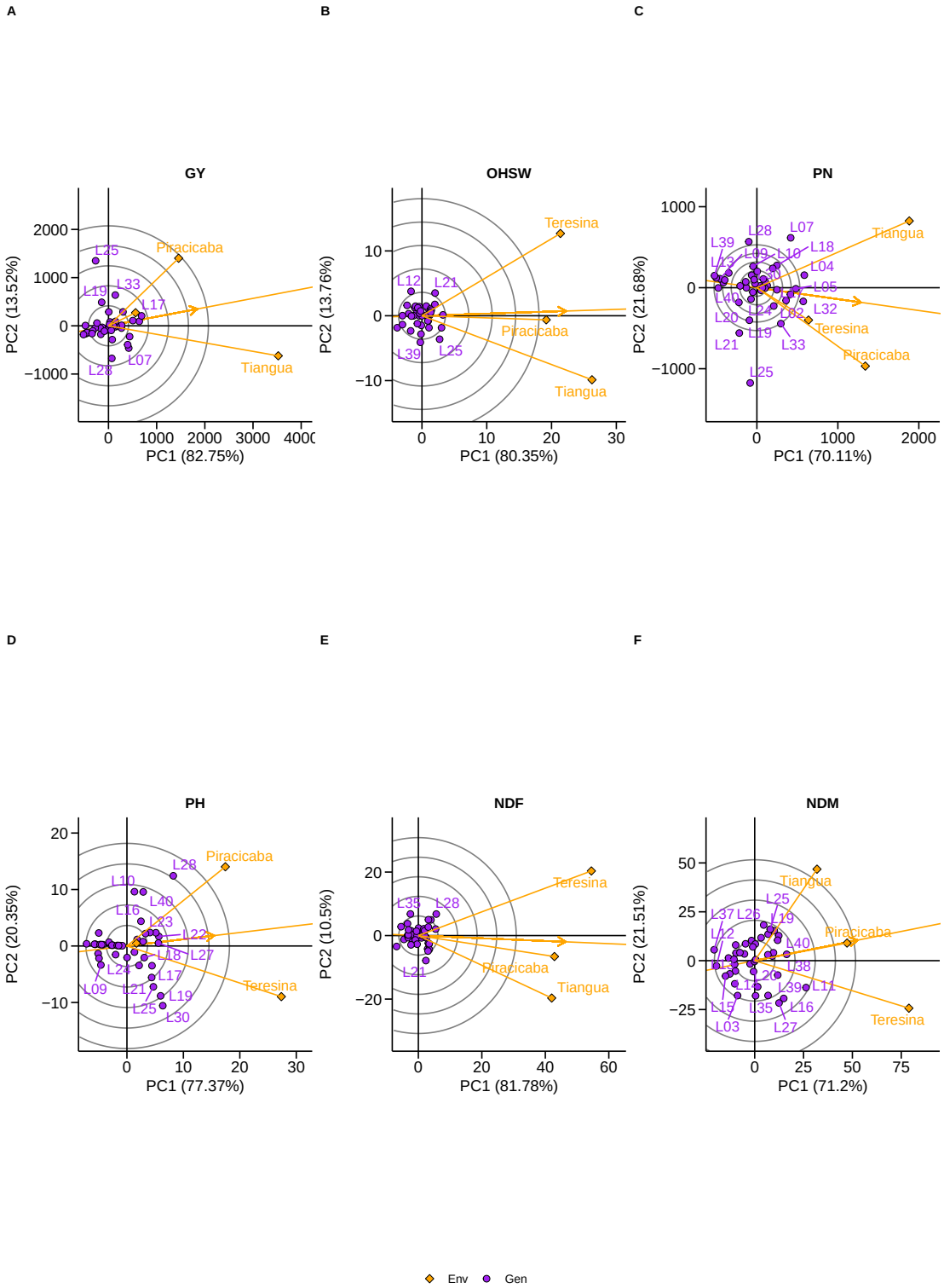

A
GY
Piracicaba
Teresina
Tiangua
PC2 (13.52%)
PC1 (82.75%)
B
OHSW
Teresina
Piracicaba
Tiangua
PC2 (13.76%)
PC1 (80.35%)
C
PN
Tiangua
Teresina
Piracicaba
PC2 (21.68%)
PC1 (70.11%)
D
PH
Piracicaba
Tiangua
Teresina
PC2 (20.35%)
PC1 (77.37%)
E
NDF
Teresina
Piracicaba
Tiangua
PC2 (10.5%)
PC1 (81.78%)
F
NDM
Tiangua
Piracicaba
Teresina
PC2 (21.51%)
PC1 (71.2%)

```
ggsave("Mean Performance vs Stability_LB.pdf", mps_panel, width = 10, height = 14, dpi = 600)
```

Discriminativeness vs. representativeness

```
library(ggplot2)
library(ggpubr)

dir.create("Discriminativeness_vs_Representativeness", showWarnings = FALSE)

dr_list <- list()

for(tr in traits){

  gge4_tr <- gge(
    LB,
    Env,
    GEN,
    resp = tr,
    centering = "environment",
    scaling = "none",
    svp = "environment"
  )
  p4 <- plot(gge4_tr,
             type = 4,
             col.env = "orange",
             col.gen = "purple",
             size.text.gen = 4,
             size.text.env = 4 ,
             repel = TRUE,
             repulsion = 2,
             col.line = "orange",
             size.line = 0.5,
             col.circle = "black",
             plot_theme = theme_metan_minimal())
  p_try <- try({
    p4_mod <- p4 +
      labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
    p4 <- p4_mod
    NULL
  }, silent = TRUE)
```

```r
  if(inherits(p_try, "try-error") || !inherits(p4, "ggplot")) {
    p_grob <- ggpubr::as_ggplot(p4)
    p4 <- p_grob +
      labs(title = tr, subtitle = NULL) +
      theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  }
  dr_list[[tr]] <- p4
}
dr_panel <- ggarrange(
  dr_list[["GY"]],  dr_list[["OHSW"]], dr_list[["PN"]],
  dr_list[["PH"]],  dr_list[["NDF"]],  dr_list[["NDM"]],
  ncol = 3, nrow = 2,
  common.legend = TRUE, legend = "bottom",
  labels = c("A","B","C","D","E","F"),
  font.label = list(size = 10, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5

)

dr_panel = annotate_figure(
  dr_panel,
  top = text_grob("", size = 14),
  fig.lab = "",
  fig.lab.pos = "bottom"
)
dr_panel
```


A
GY
PC2 (13.52%)
PC1 (82.75%)
Piracicaba
Tiangua
B
OHSW
PC2 (13.76%)
PC1 (80.35%)
Teresina
Piracicaba
Tiangua
C
PN
PC2 (21.68%)
PC1 (70.11%)
Tiangua
Teresina
Piracicaba
D
PH
PC2 (20.35%)
PC1 (77.37%)
Piracicaba
Teresina
E
NDF
PC2 (10.5%)
PC1 (81.78%)
Teresina
Piracicaba
Tiangua
F
NDM
PC2 (21.51%)
PC1 (71.2%)
Tiangua
Piracicaba
Teresina
Env
Gen

```r
ggsave("Discriminativeness vs Representativeness_LB.pdf", dr_panel, width = 10, height = 14,
```

Desire index

```r
# DESIRE INDEX FUNCTION
gainDiff <- function(ebvMat, b_ebv, dGain, nSel){

  take <- order(ebvMat %*% b_ebv, decreasing = TRUE)[1:nSel]

  mu_Sel <- colMeans(ebvMat[take, ])
  resp   <- mu_Sel - colMeans(ebvMat)
  resp   <- resp / sqrt(sum(resp^2))

  sum((resp - dGain)^2)
}
```

```r
ebvMat <- ebvMat[complete.cases(ebvMat), ]
#COV MATRIX -EBVS
library(AlphaSimR)

G_ebv <- AlphaSimR::popVar(ebvMat)
G_ebv
```

```
           [,1]       [,2]       [,3]       [,4]       [,5]       [,6]
[1,] 6587.97833 217.258861 424.765720 256.943342 36.6766871 4080.22632
[2,]  217.25886   7.323476  14.181373   8.521450  1.2173849  134.59974
[3,]  424.76572  14.181373  28.804036  17.134163  2.4397036  266.63926
[4,]  256.94334   8.521450  17.134163  10.367730  1.4703171  160.61068
[5,]   36.67669   1.217385   2.439704   1.470317  0.2129143   22.92692
[6,] 4080.22632 134.599743 266.639262 160.610683 22.9269219 2564.70381
```

```r
#desired gains
desiredGain <- c(
  GY   =  2,
  OHSW =  0,
  NDF  = -1,
  NDM  = -1,
  PH   =  0,
  PN   =  1
```

```r
)
desiredGain <- desiredGain / sqrt(sum(desiredGain^2))
sum(desiredGain^2)
```

```
[1] 1
```

```r
b_ebv <- solve(G_ebv) %*% desiredGain
b_ebv
```

```
            [,1]
[1,]  0.02346850
[2,] -0.05999296
[3,]  0.24661418
[4,] -2.78876486
[5,]  9.45275224
[6,]  0.03046048
```

```r
nSel <- 5
opt <- optim(
  par     = b_ebv,
  fn      = gainDiff,
  ebvMat  = ebvMat,
  dGain   = desiredGain,
  nSel    = nSel
)
opt_weights <- opt$par
opt_weights
```

```
            [,1]
[1,]  0.02346850
[2,] -0.05999296
[3,]  0.24661418
[4,] -2.78876486
[5,]  9.45275224
[6,]  0.03046048
```

```r
DesireIndex <- ebvMat %*% opt_weights

DesireIndex <- data.frame(
  GEN = rownames(ebvMat),
```

```r
  DesireIndex = as.numeric(DesireIndex)
) |>
  arrange(desc(DesireIndex))

head(DesireIndex, 10)
```

```
   GEN DesireIndex
1  L06    203.1275
2  L27    202.8568
3  L14    202.5262
4  L21    202.3542
5  L34    202.3358
6  L28    202.1624
7  L19    202.1393
8  L09    201.9950
9  L13    201.9429
10 L35    201.8807
```

```r
write.xlsx(DesireIndex, "Desireindex.xlsx")
```

```r
selected <- DesireIndex[1:5, ]
selected
```

```
  GEN DesireIndex
1 L06    203.1275
2 L27    202.8568
3 L14    202.5262
4 L21    202.3542
5 L34    202.3358
```

```r
sel_mat <- ebvMat[selected$GEN, ]

expResp <- colMeans(sel_mat) - colMeans(ebvMat)
scaleExpResp <- expResp / sqrt(sum(expResp^2))

df_vect = cbind(
  Desired  = desiredGain,
  Achieved = scaleExpResp
)
```

```r
ggplot(df_vec, aes(x = Desired, y = Achieved)) +
  geom_point(size = 3, color = "purple") +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  geom_text(aes(label = Trait), nudge_x = 0.05, size = 5, color = "orange") +
  coord_equal(xlim = c(-1,1), ylim = c(-1,1)) +
  labs(
    x = "Desired Gain",
    y = "Achieved Gain"
  ) +
  theme_classic(base_size = 14)
```

```r
library(dplyr)
df <- dplyr::arrange(DesireIndex, dplyr::desc(DesireIndex)) |>
  dplyr::mutate(
    GEN = factor(GEN, levels = GEN),
    Selected = ifelse(GEN %in% selected$GEN, "Selected", "Not selected")
  )
```

```r
library(ggplot2)

desiredgainplot = ggplot(df, aes(x = GEN, y = DesireIndex, fill = Selected)) +
  geom_col(width = 0.9, color = "black", size = 0.2) +
  coord_polar() +
  scale_fill_manual(
    values = c(
      "Selected"     = "#6A1B9A",
      "Not selected" = "orange"
    )
  ) +
  labs(face = "bold",
    fill  = NULL
  ) +
  theme_minimal(base_size = 14) +
  theme(
    axis.text.x = element_text(face = "bold"),
    axis.title = element_blank(),
    axis.text.y = element_blank(),
    axis.ticks = element_blank(),
    panel.grid = element_blank(),
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = "bottom"
  )
```

```r
ggsave("desiredgainplot_LB.pdf", plot = desiredgainplot,width = 10, height = 8,dpi = 600)
```

Table Selection Gain

```r
h2_df <- gmd(mixedmodel, "h2")
h2 <- h2_df$h2
names(h2) <- h2_df$VAR
traits_DI <- colnames(ebvMat)
h2_DI <- h2[traits_DI]
library(dplyr)

genetic_gain_table <- function(ebvMat, selected_gen, h2){

  traits <- colnames(ebvMat)

  res <- lapply(traits, function(tr){

    Xo <- mean(ebvMat[, tr], na.rm = TRUE)
    Xs <- mean(ebvMat[selected_gen, tr], na.rm = TRUE)

    SD  <- Xs - Xo
    SDp <- (SD / Xo) * 100

    SG  <- SD * h2[tr]
    SGp <- (SG / Xo) * 100

    tibble(
      Trait     = tr,
      `h²` = round(h2[tr], 2),
      Xo        = round(Xo, 2),
      Xs        = round(Xs, 2),
      SD        = round(SD, 2),
      `SD (%)` = round(SDp, 2),
      SG        = round(SG, 2),
      `SG (%)` = round(SGp, 2)
    )
  })

  bind_rows(res)
}
```

```r
Table3_DesireIndex <- genetic_gain_table(
  ebvMat = ebvMat,
  selected_gen = selected$GEN,
  h2 = h2_DI
)

Table3_DesireIndex
```

```
# A tibble: 6 x 8
  Trait  `h²`    Xo    Xs    SD `SD (%)`    SG `SG (%)`
  <chr> <dbl> <dbl> <dbl> <dbl>    <dbl> <dbl>    <dbl>
1 GY     0.36 749.  783.  34.3      4.59 12.4      1.65
2 OHSW   0.81  32.2  33.7  1.51     4.69  1.22     3.78
3 NDF    0.81  43.6  46.0  2.37     5.44  1.91     4.39
4 NDM    0.47  87.8  88.8  1.01     1.15  0.48     0.55
5 PH     0.23  42.5  42.8  0.26     0.62  0.06     0.14
6 PN     0.34 575.  594.  18.6      3.23  6.24     1.09
```

```r
write.xlsx(Table3_DesireIndex, "Table3_DesireIndex.xlsx", rowNames = FALSE)
```

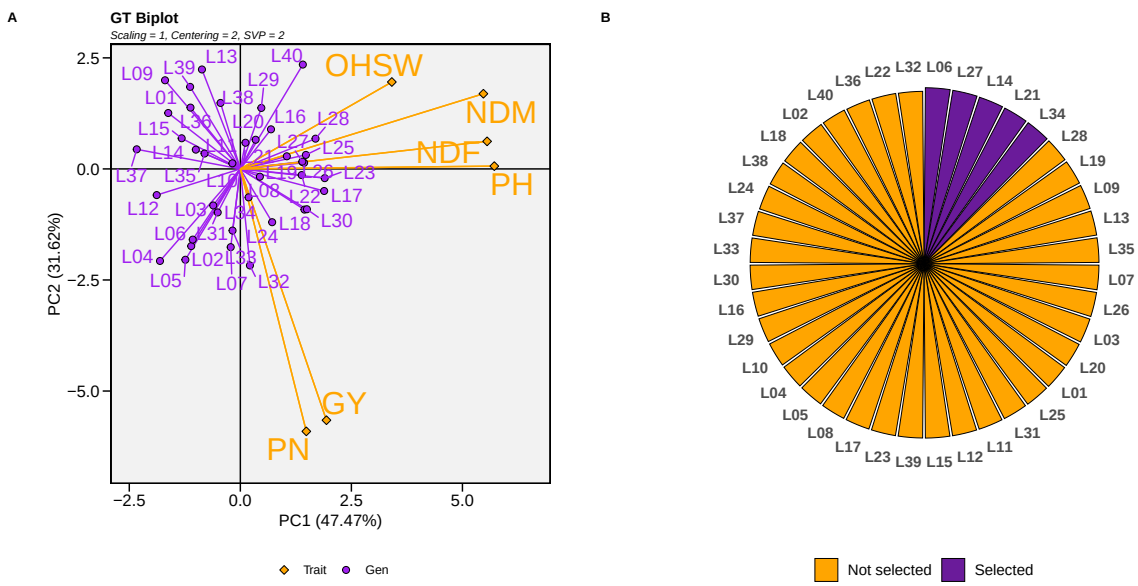
Genotype by trait biplot

```r
gtb1 = gtb(LB, GEN, resp = c(NDF, NDM, PH, PN, GY, OHSW))

gtb1plot = plot(gtb1,
     col.env = "orange",
            col.gen = "purple",
            size.text.gen = 5,
            size.text.env = 8, repel = T,
            repulsion = 2,
            col.line = "orange",
            plot_theme = theme_metan())
```

Panel Desire Gain Index + GYT Biplot

```r
gt_desired_panel <- ggarrange(gtb1plot, desiredgainplot,
  ncol = 2,
```

```
  common.legend = F, legend = "bottom",
  labels = c("A","B"),
  font.label = list(size = 10, face = "bold"),
  align = "hv",
  hjust = -0.5,
  vjust = 1.5
)
gt_desired_panel
```




```
ggsave("gt_desired_panel_LB.pdf", plot = gt_desired_panel, width = 12, height = 6,dpi = 600)
```

Density plot

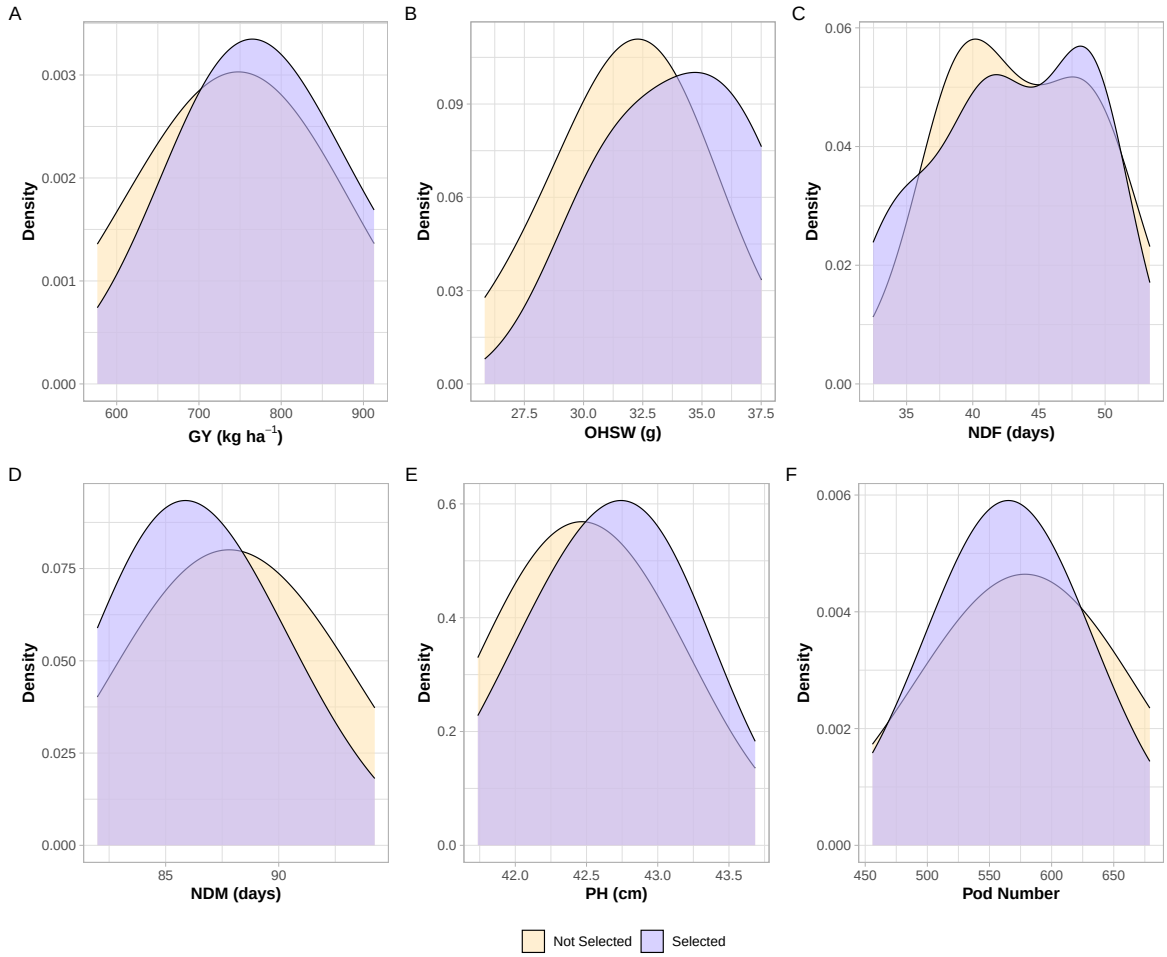
```
if ("package:plyr" %in% search()) detach("package:plyr", unload = TRUE)
library(dplyr)
library(tidyr)
library(ggplot2)
library(ggridges)
trait_names <- c(
  GY_BLUP   = "Grain Yield (GY)",
  OHSW_BLUP = "One Hundred Seed Weight (OHSW)",
  NDF_BLUP  = "Days to Flowering (NDF)",
```

```r
  NDM_BLUP  = "Days to Maturity (NDM)",
  PH_BLUP   = "Plant Height (PH)",
  PN_BLUP   = "Pod Number (PN)"
)
df_long <- bind_rows(
  lapply(names(BLUPgen), function(tr) {
    BLUPgen[[tr]] %>%
      dplyr::mutate(
        Trait  = trait_names[tr],
        Status = ifelse(GEN %in% selected$GEN,
                        "Selected", "Not Selected")
      ) %>%
      select(GEN, Trait, Status, Predicted)
  })
)
df_long$Trait <- factor(
  df_long$Trait,
  levels = rev(trait_names)
)
```

```r
plot_density_SG <- function(data_trait, selected, xlab, bw = NULL) {

  df <- data_trait %>%
    dplyr::mutate(
      Status = factor(
        ifelse(GEN %in% selected$GEN, "Selected", "Not Selected"),
        levels = c("Not Selected", "Selected")
      )
    )

  ggplot(df, aes(x = Predicted, fill = Status)) +
    geom_density(alpha = 0.6, color = "black", linewidth = 0.4, bw = bw) +

    scale_fill_manual(
      values = c("Not Selected" = "#FFE5B4",
                 "Selected"     = "#BFB4FF")
    ) +

    labs(
      x = xlab,   # nome do trait + unidade
      y = "Density",
      fill = NULL
```

```r
    ) +

    theme_light(base_size = 12) +
    theme(
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      legend.position = "bottom"
    )
}

library(patchwork)

pd1 <- plot_density_SG(BLUPgen$GY_BLUP, selected,
                       expression(bold("GY (kg ha"^{-1}*")")), bw = 100)

pd2 <- plot_density_SG(BLUPgen$OHSW_BLUP, selected,
                       expression(bold("OHSW (g)")), bw = 2.5)

pd3 <- plot_density_SG(BLUPgen$NDF_BLUP, selected,
                       expression(bold("NDF (days)")), bw = 3)

pd4 <- plot_density_SG(BLUPgen$NDM_BLUP, selected,
                       expression(bold("NDM (days)")), bw = 3.5)

pd5 <- plot_density_SG(BLUPgen$PH_BLUP, selected,
                       expression(bold("PH (cm)")), bw = 0.5)

pd6 <- plot_density_SG(BLUPgen$PN_BLUP, selected,
                       expression(bold("Pod Number")), bw = 65)
densplot = (pd1 | pd2 | pd3) /
(pd4 | pd5 | pd6) +
  plot_annotation(
    tag_levels = "A",
    theme = theme(
      plot.tag = element_text(
        face = "bold",
        size = 14
      )
    )
  ) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```

```r
densplot
```




```r
ggsave(plot = densplot, "densplot.pdf", width = 12, height = 10, dpi = 600)
```